Quality Information Document

Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

Recent Changes of the Black Sea Wave Power

Issue: 1.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
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I INTRODUCTION

I.1 Product name

OMI_EXTREME_WAVE_BLKSEA_wave_power

I.2 Indicator evaluation method

The Wave Power P is defined by:

$$P = \frac{\rho g^2}{64\pi} H_s^2 T_e$$

Where ρ is the surface water density, g the acceleration due to gravity, Hs the significant wave height (VHM0), and Te the wave energy period (VTM10) also abbreviated with Tm-10. The extreme statistics and related recent changes are defined by (1) the 99th percentile of the Wave Power, (2) the linear trend of 99th percentile of the Wave Power, and (3) the difference (anomaly) of the 99th percentile of the last available year in the multiyear dataset compared against the long-term average. The statistics are based on the period 1950 to -1Y and are obtained from yearly averages.

I.3 CMEMS products used as input for the computations

Product name (and Type)	Documentation
BLKSEA_MULTIYEAR_WAV_007_006	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BLK-QUID-007-006.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-BLK-PUM-007-006.pdf

II SCIENTIFIC VALIDATION

II.1 Black Sea extreme wave events

More details and full scientific evaluation can be found in the CMEMS Ocean State Report issue 6 (von Schuckmann et al., 2022), Sect. 2.8.

As the variables of this OMI cannot be validated directly, selected validations of the significant wave height (SWH) are shown. The SWH and wave energy period of the product BLKSEA_MULTIYEAR_ WAV_007_006 are the basis of the wave power computations and a thorough validation of these parameters is supposed to provide reliable results also for the wave power parameters.

The most extensive validation of SWH was done by comparing L3 altimetry satellite data with the model data interpolated to the satellite data (Figure 1). It is important to mention that the data of these satellite, namely Sentinel-3a/-3b, Cryosat-2, CFOSat, Saral/Altika, and Envisat, has not been assimilation as done for the Jason missions. Reassuringly, the model skill is nearly identical to the values obtained for the assimilated satellite data. Root mean square difference (RMSD) is of the same magnitude (23 cm), scatter index (SI) is 2% worse, overall bias is now negative (-4 cm), and the correlation is 0.01 lower. As a result of the negative bias, the slope is 0.05 lower. Crucial for the OMI quality is also the model performance of the high wave heights as they contain much more energy, which show no obvious tendency to be over- or underestimated and the scatter around the diagonal is rather low.

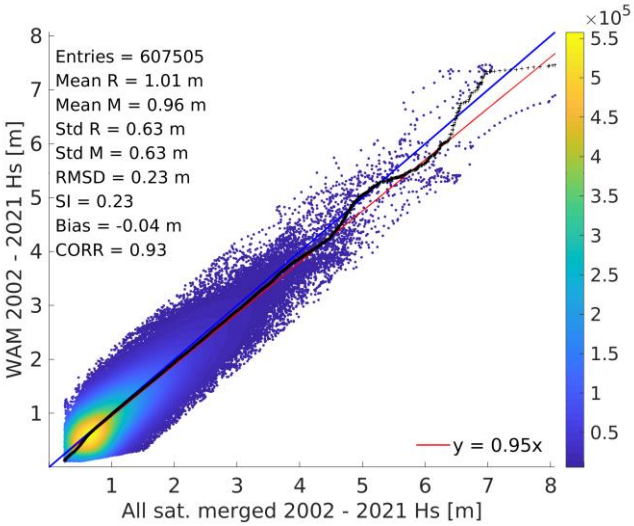

Figure 1: *Scatter plot showing satellite measurements versus modelled significant wave heights (Hs) for the period 2002-2021 using the non-assimilated satellites Sentinel-3a/-3b (NRT), HaiYang-2b (NRT), Cryosat-2 (NRT+MY), CFOSat (NRT+MY), Saral/Altika (NRT+MY), and Envisat (MY) (data from individual satellites was merged). Also shown are the estimated bivariate probability density (colours), the least squares linear fit of modelled vs observed wave heights (red line), specific quantiles taken from the empirical cumulative density function (black line), and the 45° diagonal (blue line). Furthermore, summary statistics and skill scores are included. R: reference (satellite) data, M: model data, SI: scatter index.*

The temporal evolution of the SWH quality is also of interest for the OMI. For this purpose, annual validations are shown in Figure 2. It shows time series of different statistical parameters for the complete validation period between 2002 and 2021 (using an annual scale). The metrics show that the skill that can be seen in all of the statistical parameters is versatile for the individual satellites. In general, the following observations can be made: During the assimilation periods of the individual satellites, the corresponding metrics are relatively constant (RMSD: 18 cm, bias: 5 cm, correlation: 0.97, SI: 0.17, and reduction of variance (RV): 0.92), whereas during those periods without data assimilation the quality of the statistics is slightly decreased. For example, RMSD increases to ~30 cm, bias decreases to -5 cm, correlation to 0.88, SI increases to 0.30, and RV decreases to 0.75. Nevertheless, even those metrics can still be considered as indicating good model performance.

Overall, the skill scores depend on the number of collocated measurements to some extent. It is obvious that with a decreasing number of observations, parameters such as RMSD, bias, and scatter index increase, while RV and correlation decrease.

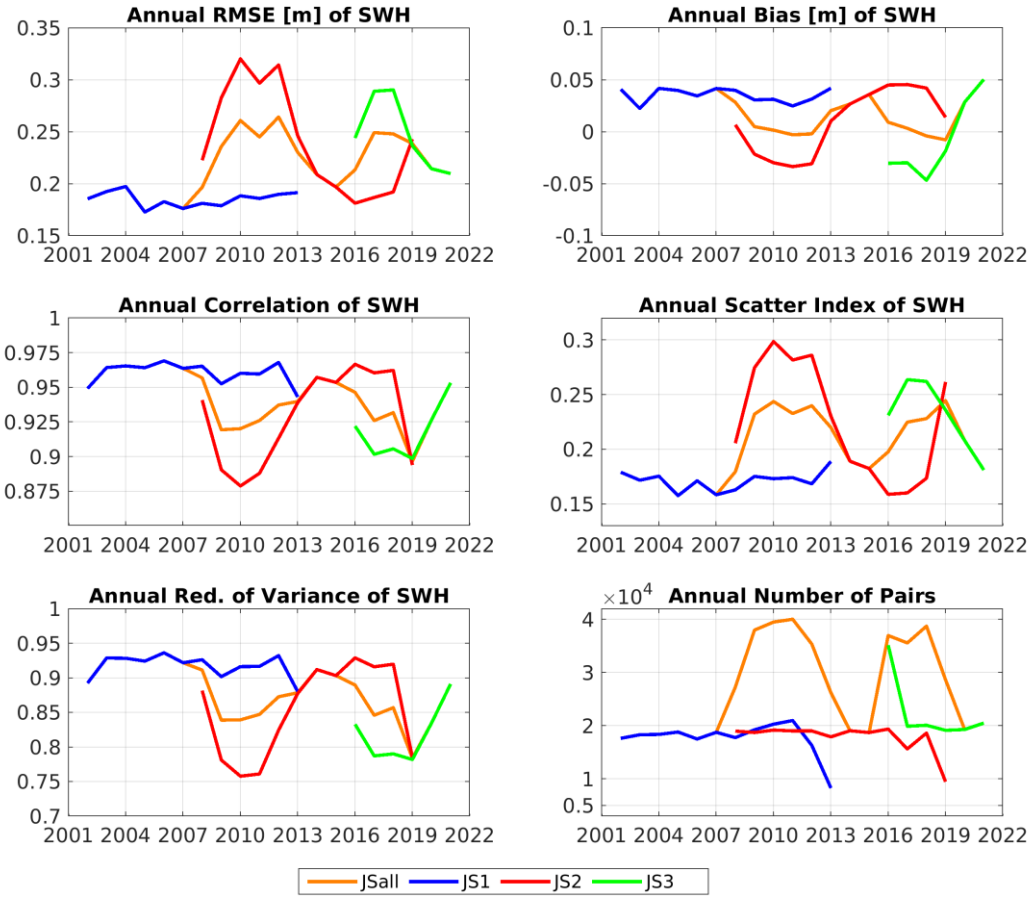

Fig. 2: *Yearly values of significant wave height (SWH) metrics for the Black Sea derived from individual and combined satellite measurements. "JS1/2/3": Jason-1/2/3 mission, "JSall": merged Jason data.*

Besides SWH, the wave energy period should be validated. However, wave buoys usually do not provide this parameter. However, it could be beneficial to have a look at the Tm02 period, which is of course not the same variable but both are often called mean wave period and so it could give a first indirect insight into the wave energy period quality. Wave Tm02 period time series (Figure 3) show a similar behaviour as SWH. The timing is very good over the whole period but a slight overall underestimation is present. However, this underestimation is mostly below 1 s, thus we can consider that the validations show good agreement with observations for the Tm02 period. The overall bias is -0.32 s and the root mean square difference (RMSD) 0.88 s (Figure 4) further supporting the good quality of the model data. It has to be noted that higher periods (> ~5 s) deviate more strongly from measurements (underestimation) than shorter periods. Similar to SWH, long periods are important for the wave energy and thus, wave energy extremes could be slightly underestimated.

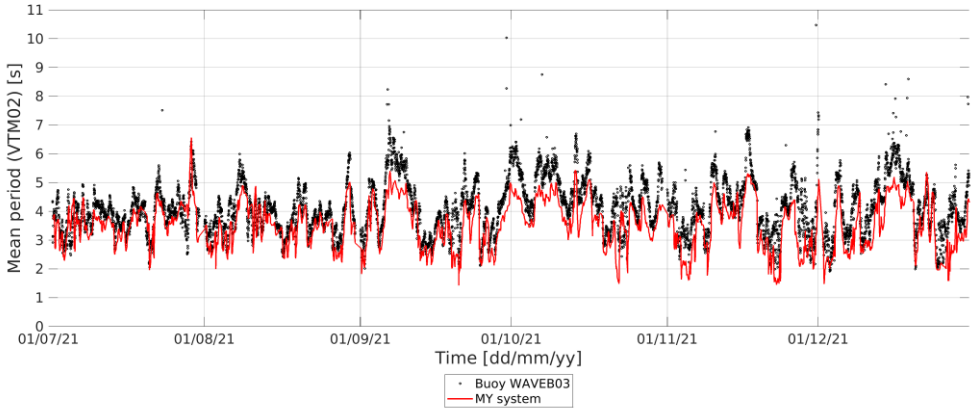

Fig. 3: Comparing modelled wave Tm02 period (VTM02) to data from the buoy WAVEB03 located at the Bulgarian coast.

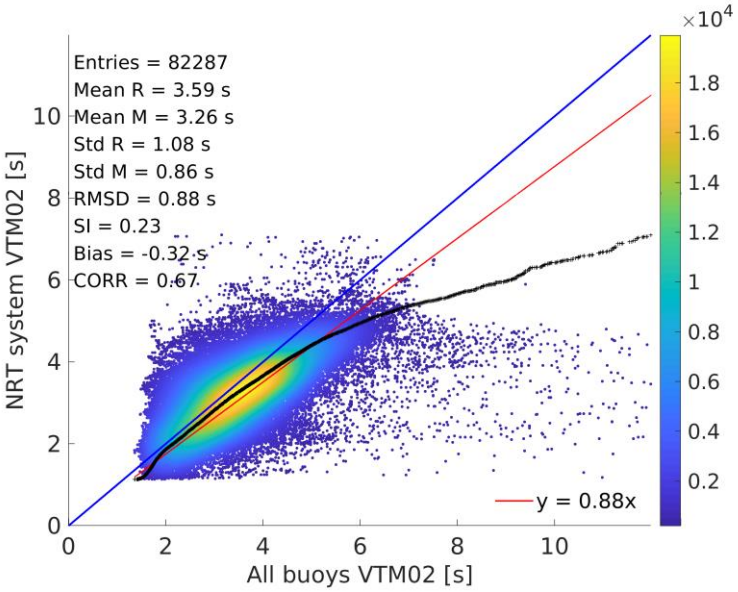

Fig. 4: Scatter plot of mean wave period (VTM02) of all available wave buoys vs. model for the period May 2020 to December 2021. Also shown are the estimated bivariate probability density (colours), the least squares linear fit of modelled vs observed wave heights (red line), specific quantiles taken from the empirical cumulative density function (black line), and the 45° diagonal (blue line). Furthermore, summary statistics and skill scores are included. R: reference (satellite) data, M: model data, SI: scatter index.

III REFERENCES

von Schuckmann, K., P.-Y. Le Traon, N. Smith, A. Pascual, S. Djavidnia, P. Brasseur, M. Grégoire (Eds.) (2022): Copernicus Ocean State Report, Issue 6, Journal of Operational Oceanography, 15:sup1, s1–s220, doi: 10.1080/1755876X.2022.2095169

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